



UNIVERSITY OF
BIRMINGHAM

Superconductivity

Lecture 07 – “Order!”

Dr Mingee Chung (Minki Jeong)

Associate Professor

Condensed Matter Physics Group

School of Physics & Astronomy

"Superconductivity" module

1. Phenomenology:
 - Experimental observations and consistency (London equations)
2. Macroscopic description:
 - Phase transition (Ginzburg-Landau model)
3. Microscopic description:
 - "Many-body" wavefunction (BCS model)

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2nd order phase transition

Constructing a free energy

Finding a minimum

Thermodynamics: entropy and heat capacity

Deep inside a bulk superconductor

- $\rho = 0$
- $B = 0$
- $E = 0$

$$\begin{aligned}\nabla \times \mathbf{j} &= -\frac{1}{\mu_0 \lambda_L^2} \mathbf{B}, \\ \frac{\partial \mathbf{j}}{\partial t} &= \frac{1}{\mu_0 \lambda_L^2} \mathbf{E},\end{aligned}$$

where

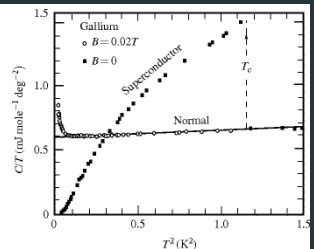
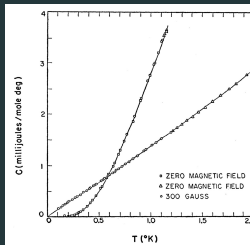
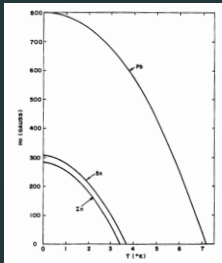
$$\frac{\mu_0 n e^2}{m_e} = \frac{1}{\lambda_L^2}.$$

Useful to suppose $n = n_n + n_s$, e.g., when explaining a sharp drop in $\rho(T)$ compared to $\lambda(T)$.

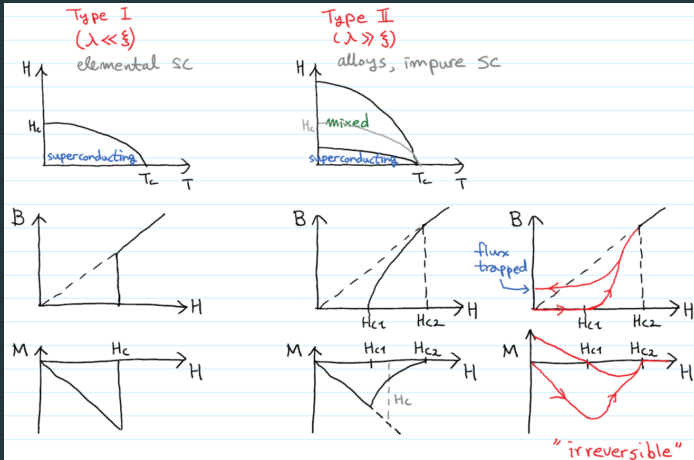
Thermodynamics (1/2)

Howe much can you say about these?

- $S_s \leq S_n$
- $H_c(T) = H_c(0) \left[1 - \left(\frac{T}{T_c} \right)^2 \right]$
- Specific heat jump at T_c



Thermodynamics (2/2)



Flux quantisation

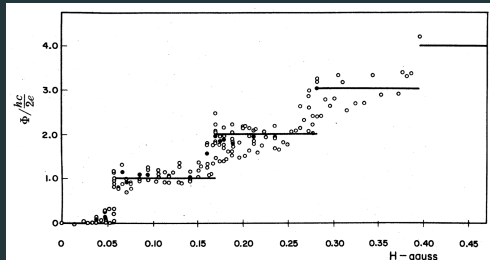


Figure 1: Quantised increase in trapped flux as a function of applied magnetic field (Deaver, 1961).

Ginzburg-Landau Theory

1. Phenomenological model
2. Focus on 2nd-order phase transitions
3. Quantitative, e.g., type-I & II distinction

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2nd order phase transition

Normal-to-superconductor

Two facts:

- T_c marks a phase transition;
- Superconducting state is more ordered,

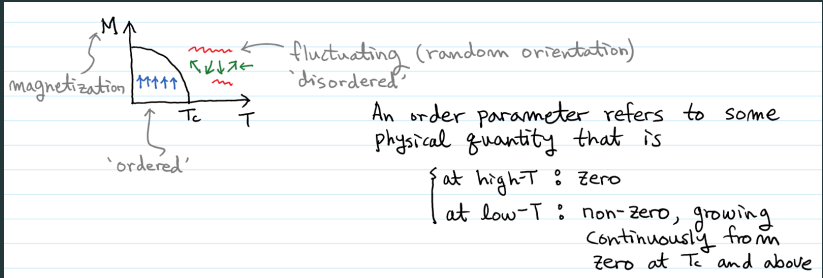
or equivalently,

- High temperature phase – 'disordered' (normal);
- Low temperature phase – 'ordered' (superconducting),

Similarity to other phenomena we know of?

⇒ a ferromagnet.

Paramagnet-to-ferromagnet

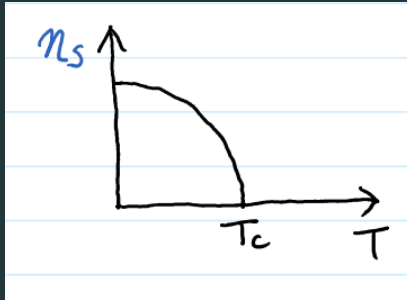


In more advanced terms,

- Ferromagnetic order = rotational symmetry broken;
- Order = broken symmetry, more generally.

Normal-to-superconductor

Compare with the superconducting electron density vs. temperature



Questions:

- Is n_s an order parameter?
- What symmetry is broken?

Constructing a free energy

A framework

Let's think of the problem

- by making analogy to familiar situation,
- on a very general ground as possible.

Recall that Max Born's interpretation of the 'wavefunction' as related to a probability density only came after the solution to the Schrodinger's wave equation for a hydrogen atom had been obtained. Ginzubrg & Landau assumed

- free energy in terms of an order parameter $\Rightarrow$ to be minimised;
- a complex order parameter $\psi = |\psi|e^{i\theta} \Rightarrow n_s = |\psi|^2$.

How to construct a free energy

Consider

$$\bullet T \simeq T_c$$

One can expand the free energy density $f(\psi)$ as

$$f_s - f_n = a_0 + a_1|\psi| + a_2|\psi|^2 + a_3|\psi|^3 + a_4|\psi|^4 + \dots, \quad (1)$$

using a Taylor expansion.

For this to have a minimum, all odd terms should go away; One can

rewrite as

$$f_s - f_n = a|\psi|^2 + \frac{1}{2}b|\psi|^4 + \dots \quad (2)$$

Some conditions (1/2)

From

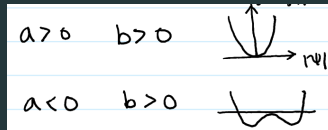
$$f_s - f_n = a|\psi|^2 + \frac{1}{2}b|\psi|^4 \quad (3)$$

one can go a bit further ...

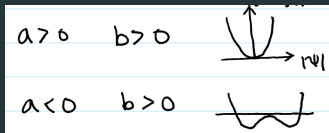
Try to implement the conditions:

- For $T > T_c$, we expect $f_s - f_n$ minimum at $|\psi| = 0$;
- For $T < T_c$, we expect $f_s - f_n$ minimum at nonzero ψ .

To satisfy the above, one is constrained for a and b as



Some conditions (2/2)



So $a(T)$ changes sign across T_c while b remains positive.

One can further constrain

$$a(T) \simeq \frac{da}{dT} (T - T_c) + \dots \quad (4)$$

$$b(T) \simeq b + \dots \quad (5)$$

since $a(T) = 0$ across T_c therefore remains very small close to T_c

Finding a minimum

Simply differentiate

$$f_s - f_n = a|\psi|^2 + \frac{1}{2}b|\psi|^4 \quad (6)$$

One obtains

$$\frac{d}{d|\psi|} (f_s - f_n) = 2a|\psi| + 2b|\psi|^3 \quad (7)$$

$$= 0. \quad (8)$$

The solutions are

- $|\psi|^2 = -a/b$ or
- $|\psi| = 0$.

One can rewrite, for $T < T_c$ and $T > T_c$, respectively,

$$|\psi| = \sqrt{\frac{1}{b} \frac{da}{dT} (T_c - T)} \quad (9)$$

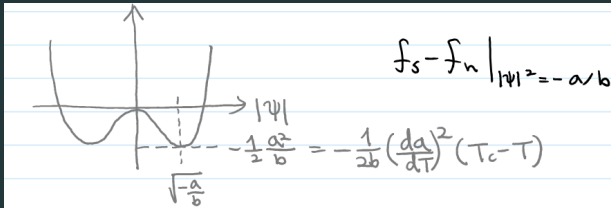
$$|\psi| = 0. \quad (10)$$

Solution

Plugging in the solution to the original equation gives

$$f_s - f_n|_{|\psi|^2 = -a/b} = -\frac{1}{2b} \left(\frac{da}{dT} \right)^2 (T_c - T)^2. \quad (11)$$

The result can be summarised in the following figure:



Thermodynamics: entropy and heat capacity

Entropy & Heat capacity

From

$$f_s - f_n|_{|\psi|^2 = -a/b} = -\frac{1}{2b} \left(\frac{da}{dT} \right)^2 (T_c - T)^2, \quad (12)$$

one can obtain

$$s_s - s_n = -\frac{d}{dT} (f_s - f_n) \quad (13)$$

$$= -\frac{1}{b} \left(\frac{da}{dT} \right)^2 (T_c - T), \quad (14)$$

and

$$c_s - c_n = T \frac{d}{dT} (s_s - s_n) \quad (15)$$

$$= T \left(\frac{1}{b} \right) \left(\frac{da}{dT} \right)^2. \quad (16)$$

What's next?

So far we have neglected ...

- $\nabla\psi(\mathbf{r})$
- a magnetic field

The former implies that we have considered a 'homogeneous' system ($\nabla\psi(\mathbf{r}) = \nabla\psi = 0$)

In the next lecture, we'll incorporate $\nabla\psi(\mathbf{r})$, or spatial inhomogeneity, into our formalism.